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EXECUTIVE SUMMARY

This project was able to achieve the complete design and analysis of a mechatronic robot gripper based on a planar four bar linkage mechanism. The final design that was developed met the main project requirements which were a 6.5 in closed gripper length, a 2.5 in maximum finger opening, compatibility with rotary actuation, and the ability to handle a 300 g object. To do this, several design concepts were made and compared, and the final mechanism was selected using a weighted evaluation process. After that, the chosen design was verified through a graphical synthesis, Grashof condition checking, and SolidWorks CAD modeling to confirm that the mechanism was compact and symmetric, as well as being capable of smooth opening and closing motion.

The project also achieved analytical and numerical validation of the mechanism. Position, velocity, acceleration, and force behaviour were studied using the selected four bar geometry. MATLAB was also used to implement the numerical model and generate the plots, while Simulink was used to observe motion trends and dynamic response. The results showed the expected behaviour of the parallelogram mechanism, where the input and output links followed the same exact angular motion while the coupler link angle remained fixed due to the mechanism geometry. Force analysis was also completed over the operating range of the gripper, which helped us evaluate the dynamic loading in the links and the torque related behaviour during motion. Overall, the project produced a full design workflow, from concept selection to CAD, kinematic analysis to dynamic analysis, and simulations to validate our findings. Even though no physical prototype was actually built, the final outcome gave us a solid and clear foundation for the future of mechatronic integration.

INTRODUCTION

The purpose of this project was to design and model, then analyze, and validate a mechatronic robotic gripper using a planar four bar linkage mechanism. The main goal was to create a gripper that gives smooth and repeatable opening and closing motion can be driven by rotary actuation, is simple enough to analyze properly and can grip a 300 g object while meeting the required geometry of 6.5 in total length when closed and 2.5 in finger spacing when fully open. In simple terms, the project was about taking a real design problem and solving it using engineering math, mechanism design, CAD and simulation tools.

The literature survey showed why a four bar linkage was a good choice for this project. From the background study, four bar linkages are widely used because they are simple, flexible, provide repeatable motion, and can convert rotary input motion into useful output motion. They are also easier to model mathematically than more complicated gripper geometries. The review also showed that important ideas such as transmission angle, Grashof's condition, joint placement, coupler motion, and force transfer all matter when designing a gripper that must move in a controlled way and still hold a load properly. This gave a solid base for choosing and studying the final mechanism.

The project problem was to design a mechanically efficient gripper that is compact and reliable, as well as analytically tractable, and easy to integrate into a mechatronic system. The main constraints were the use of a planar four bar linkage, a 3 second operating cycle, rotary actuation compatibility, and limited project time and computational resources. Based on this the project objectives were to model the mechanism using engineering mathematics and kinematics, use graphical synthesis and Grashof's condition to ensure feasible motion, develop a SolidWorks CAD assembly, carry out analytical position, velocity, and acceleration analysis, implement MATLAB simulations, compare results from more than one method, and present the work using proper engineering documentation. The scope of the work included concept generation, graphical synthesis, mobility verification, CAD modeling, kinematic equation development, force analysis, MATLAB simulation, comparison of results, and discussion of mechatronic integration. At the same time, the project had a few limits, since no physical prototype was built, actuator sizing and closed loop control were only discussed conceptually, friction and manufacturing effects were neglected, and sensor integration was not validated experimentally.

The expected outcome of the project was to show proper use of engineering fundamentals on a realistic design problem and to demonstrate the ability to model and analyze, then validate, and clearly communicate a mechanical system using modern tools. The course schedule also shaped the work. The project moved from problem definition, literature review, and requirement specification, into mechanism selection and graphical synthesis, then CAD modeling, analytical

and force analysis, MATLAB implementation, results interpretation, and finally final report writing. So overall, this project was not just about making a gripper shape. It was really about following a full engineering design process from start to finish, which I think made the whole project more useful and more realistic too.

ENGINEERING DESIGN METHODOLOGY

Mechanism Synthesis

Graphical Synthesis

(Refer to [Appendix A.1 - Graphical Synthesis Calculations](#))

To construct the graphics synthesis for our design, we had to take the different constraints into consideration. First off, when the claw is fully open, it must be 2.5 inches in length apart from one finger to another. Next, when the claw is fully closed, it must have a total length of reach of 6.5 inches. To address both of these constraints, we had to derive the open and closed configurations by determining the link lengths and pivot spacing, as well as some angles for rotation. Collectively, we decided that:

$$l_1 = 2 \text{ inches}$$

$$l_2 = l_4 = 3 \text{ inches}$$

$$l_3 = 2 \text{ inches (+ 1.5 inches to satisfy criteria)}$$

The gap between the two single claws is 0.75 inches, and so accordingly, the distance from each origin to the end of the claw finger must be 0.875 inches to obtain a distance of 2.5 inches between each claw finger when fully open, respectively. Also, the angle when each claw is fully open can be found by simple trigonometry/ratios, where $\theta_2 = 16.96 \text{ degrees}$. When the claw is fully closed, the value of $\theta_2 = 0$. Also, to achieve a max reach distance of 6.5 inches, when fully closed, the respective links $l_2 + l_3 + 1.5 = 3 + 2 + 1.5 = 6.5 \text{ inches}$.

Grashof Condition Verification

The shortest link S in the design is 2in and the longest link L is 3in, the remaining 2 links P & Q are 2in and 3in. Applying the Grashof's criterion:

$$S + L \leq P + Q$$

$$2 + 3 \leq 2 + 3$$

$$5 \leq 5$$

$$5 = 5$$

Since $S + L = P + Q$, the mechanism satisfies Grashof's criterion and can be therefore classified as a Grashof mechanism, meaning at least one link can completely rotate relative to the other links.

Mechanism Classification

Since $S + L = P + Q$, the mechanism is of the *change point mechanism* class. Because $l_1 = l_3$ and $l_2 = l_4$, the mechanism can be referred to as the special case of a change point mechanism called a *parallelogram four-bar mechanism*.

Since S can be referred to as the base link, the motion type or the mechanism behaviour can be classified as a *crank rocker mechanism*.

CAD Modeling and Assembly

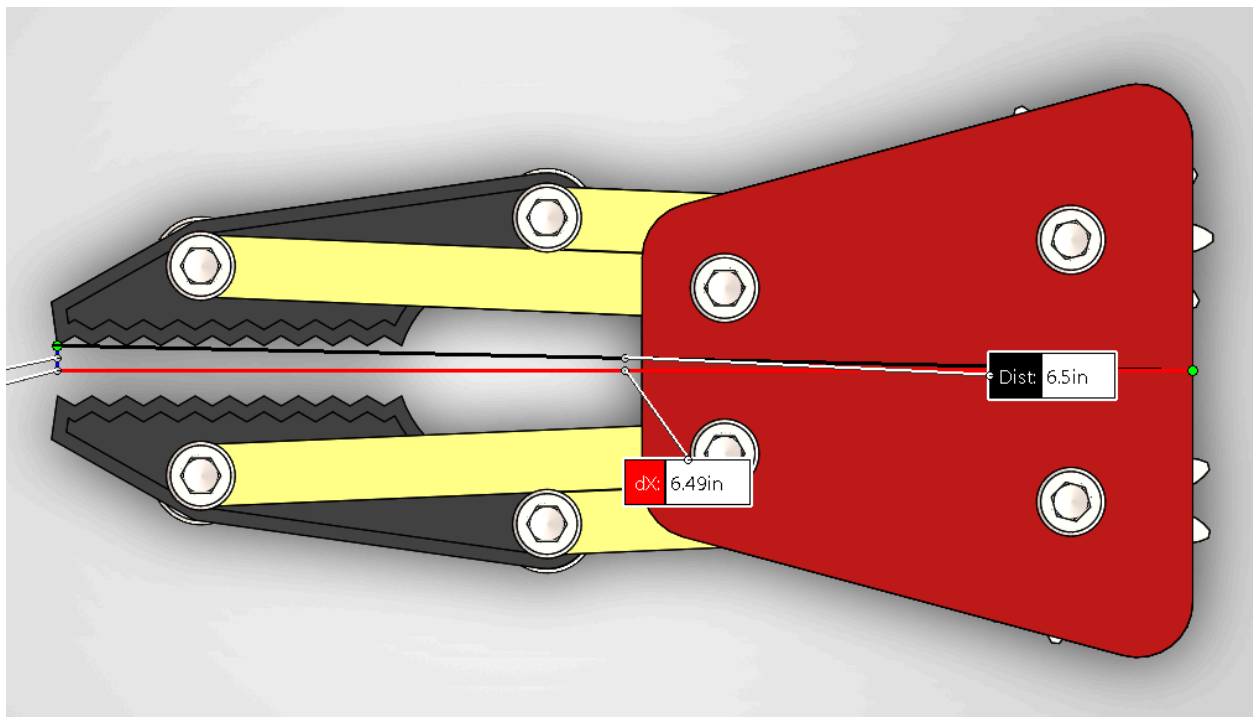


Figure 1. Closed Claw Mechanism Total Length

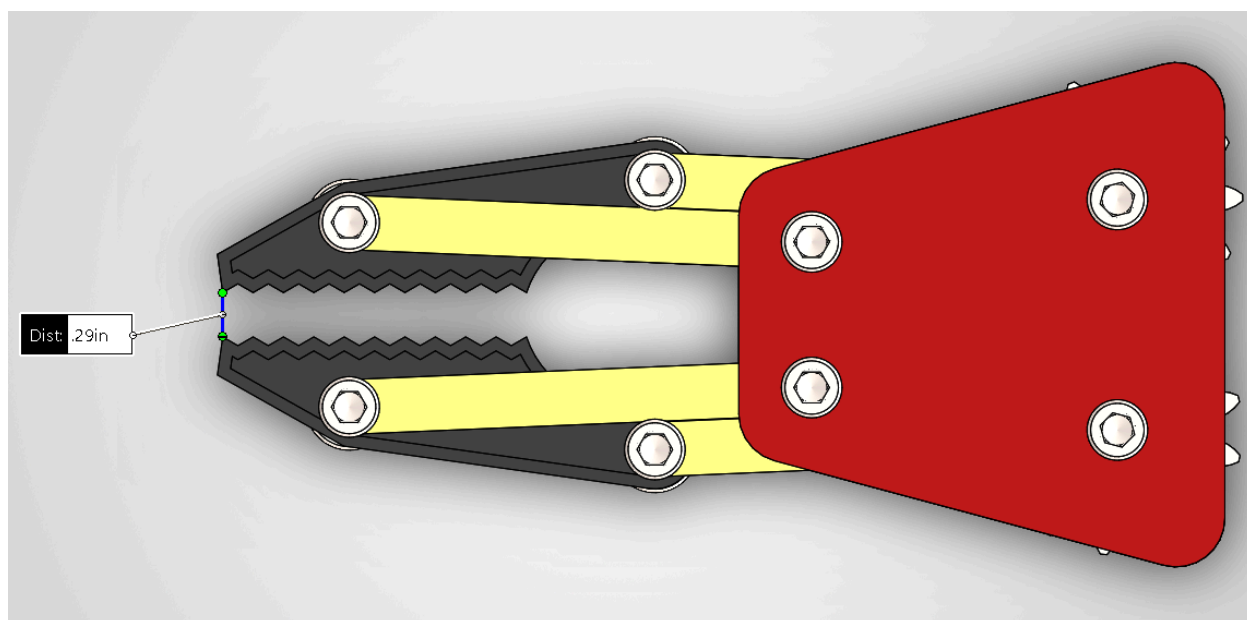


Figure 2. Closed Claw Mechanism Gripper Distance

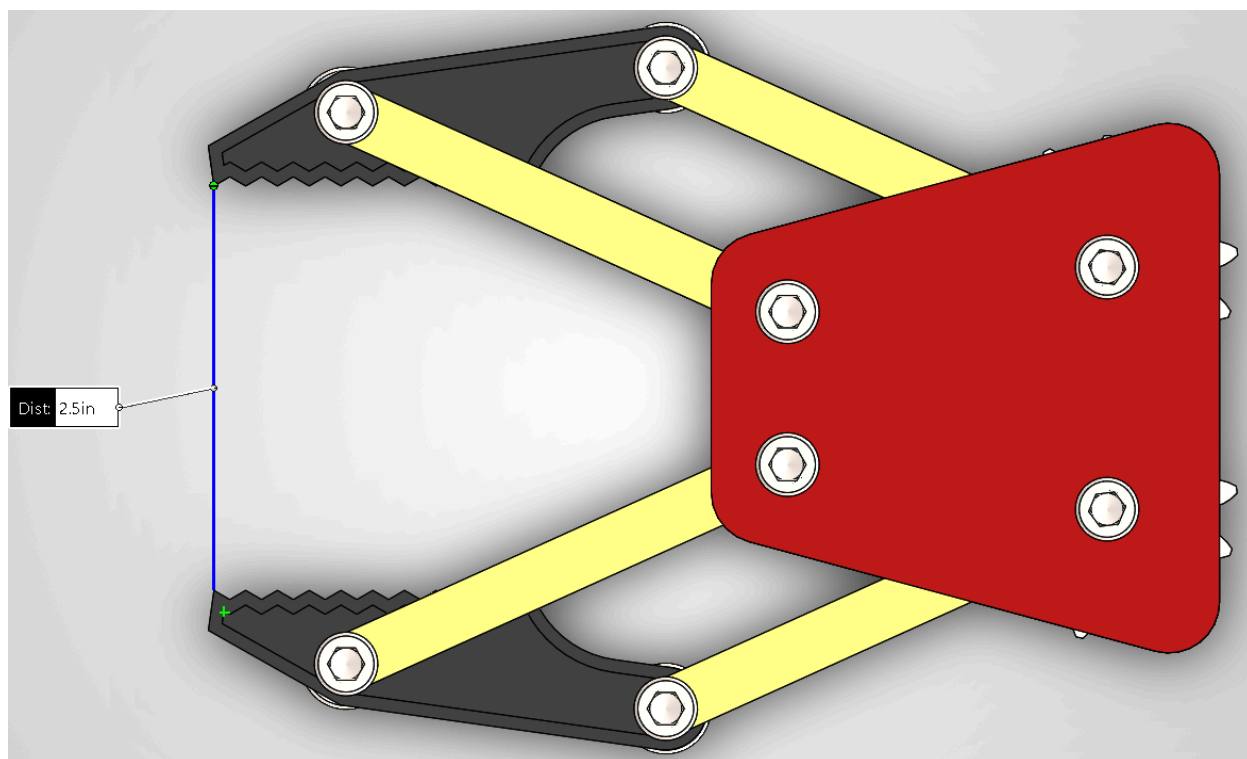


Figure 3. Open Claw Mechanism Gripper Distance

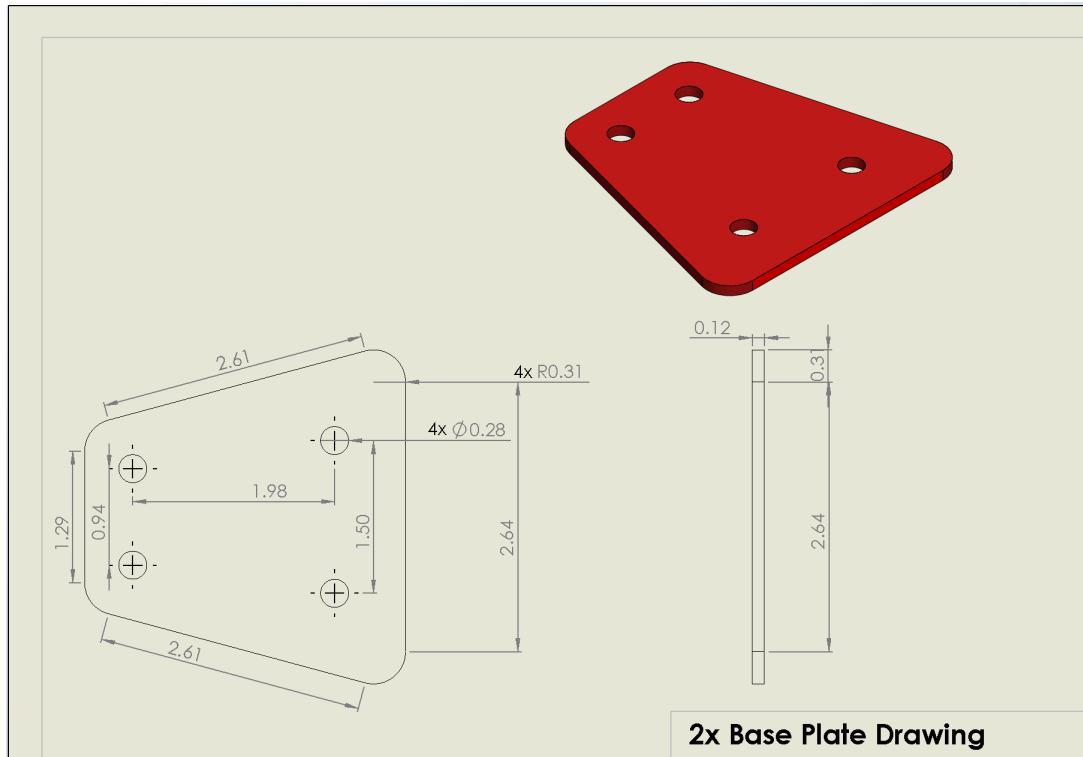


Figure 4. SolidWorks Drawing of Base plate (1:1)

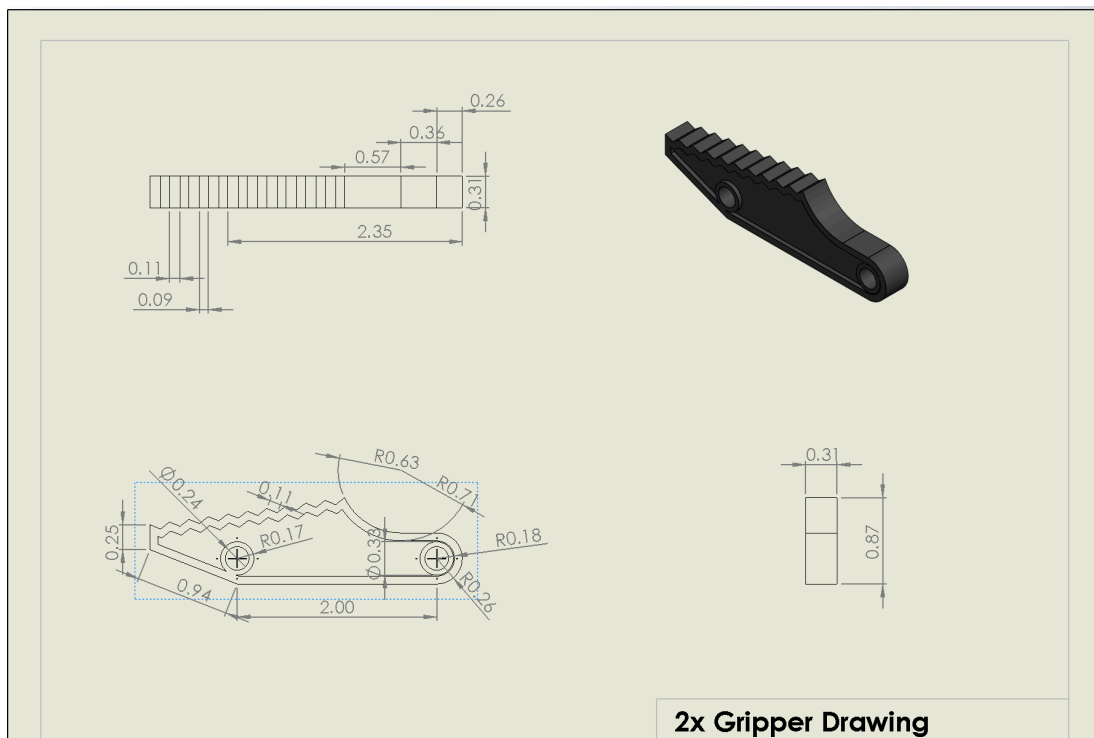


Figure 5. SolidWorks Drawing of Gripper (1:1)

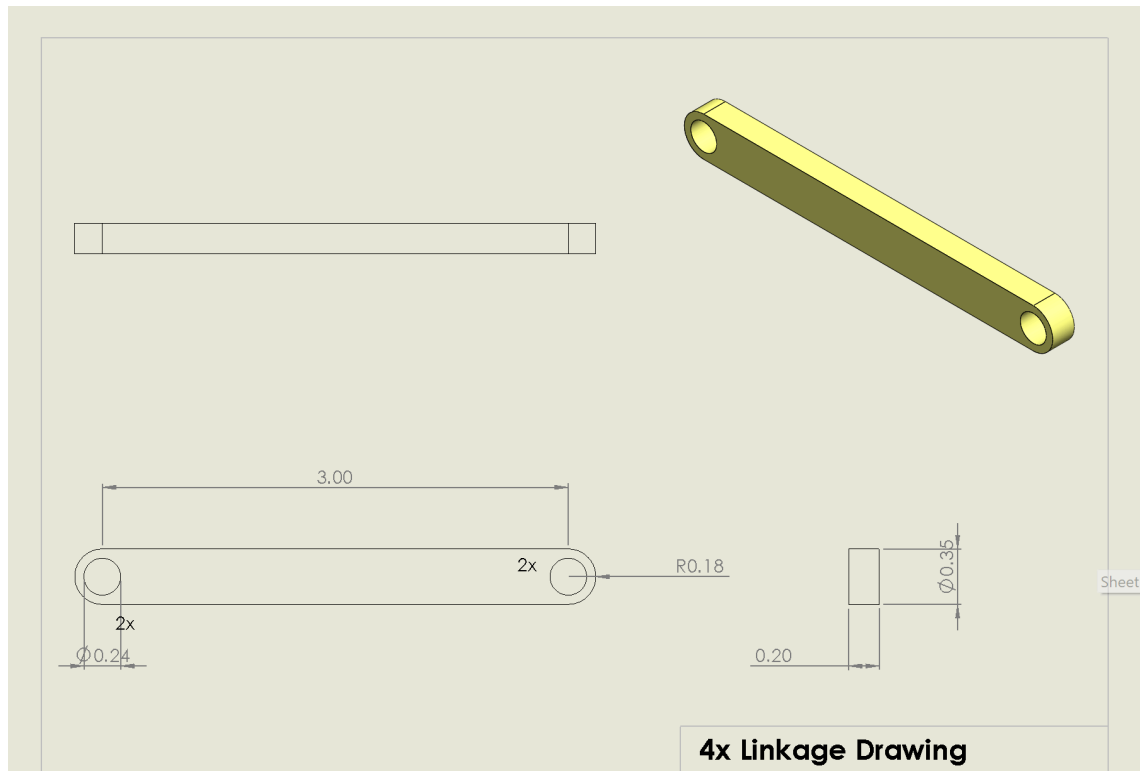


Figure 6. SolidWorks Drawing of Linkages (1:1)

KINEMATIC & DYNAMIC ANALYSES AND VALIDATION

Assumptions

1. Friction is ignored in the system
2. Only one side of gripper analyzed due to symmetry
3. Global coordinate system used
4. Ground link inclined by 7mm offset (O_4 is 7mm higher than O_2)
 - a. So $\theta_1 = \theta_3 = 7.92^\circ$
5. Planar rigid links used for analysis (infinitely stiff)
6. Ideal pin joints used for analysis

Position Analysis

(Refer to [Appendix A.2. Assembly Configuration and Setup](#))

If we used the vector-loop method, the closed-loop equation is defined as,

$$R_2 + R_3 - R_4 - R_1 = 0$$

We can substitute the Euler equivalent of each term,

$$\begin{aligned} ae^{j\theta_2} + be^{j\theta_3} - ce^{j\theta_4} - de^{j\theta_1} &= 0 \\ 3e^{j\theta_2} + 2e^{j\theta_3} - 3e^{j\theta_4} - 2e^{j\theta_1} &= 0 \end{aligned}$$

Because our mechanism is a special case of a parallelogram four bar mechanism, $\theta_2 = \theta_4$ and $\theta_1 = \theta_3$, we can replace them as θ_{24} and θ_{13}

$$3e^{j\theta_{24}} + 2e^{j\theta_{13}} - 3e^{j\theta_{24}} - 2e^{j\theta_{13}} = 0$$

All terms cancel out, and we are left with,

$$0 = 0$$

Therefore for this special case, the vector-loop equations can be reduced and simplified significantly to perform the mechanism analysis.

From [Appendix A.2](#) we know that the input crank and output rocker are the same length ($l_2 = l_4$) and the ground link and coupler are also the same length ($l_1 = l_3$). Therefore, the following relations must be true,

$$\theta_2 = \theta_4 \quad (2.1)$$

and

$$\theta_1 = \theta_3$$

We can find θ_1 through basic trigonometry that the distance between O_2 and O_4 as well as the vertical distance between O_2 and O_4 are known,

$$\theta_1 = \sin^{-1}\left(\frac{7mm}{50.8mm}\right) = 7.92^\circ$$

Therefore,

$$\theta_1 = \theta_3 = 7.92^\circ = 0.138rad \quad (2.2)$$

Velocity Analysis

We can differentiate Equations 2.1 and 2.2 to obtain expressions for the angular velocity.

$$\omega_2 = \omega_4 \quad (2.3)$$

$$\omega_1 = \omega_3 = 0 \quad (2.4)$$

Acceleration Analysis

We can further differentiate equation 2.3 and equation 2.4 to determine the expressions for angular acceleration.

$$\alpha_2 = \alpha_4 \quad (2.5)$$

$$\alpha_1 = \alpha_3 = 0 \quad (2.6)$$

Input Angle Range Calculation

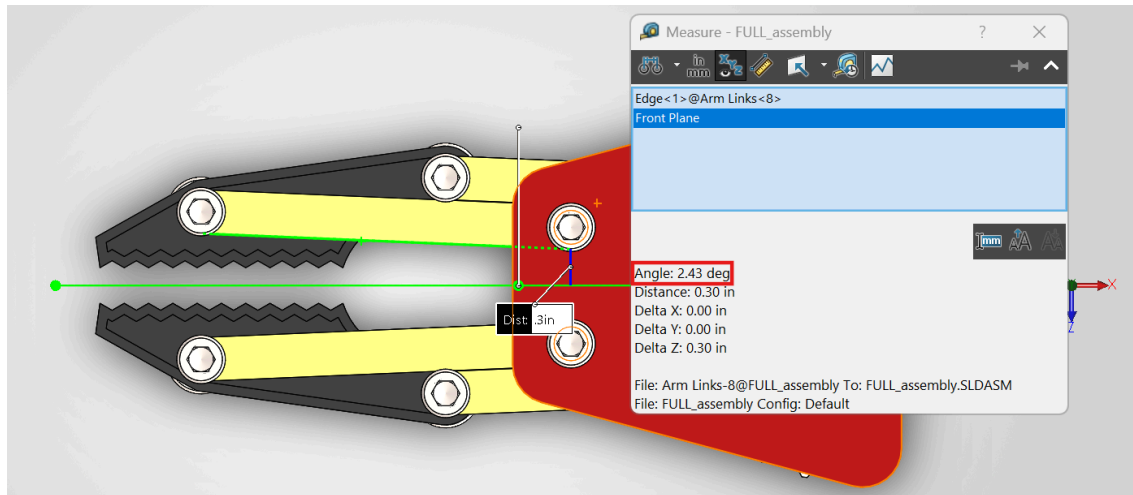


Figure 7. Closed Claw Mechanism with Link 2 Angle

From Figure 7, we can see that when the mechanism is fully closed, link 2 makes a 2.43 degree angle with the *negative* global X axis.

The angle relative to the *positive* global X axis ($\theta_{2_{max}}$) can be determined,

$$\theta_{2_{max}} = 180^\circ - 2.43^\circ$$

$$\theta_{2_{max}} = 177.57^\circ$$

$$\text{Rounding} \rightarrow \theta_{2_{max}} = 178^\circ$$

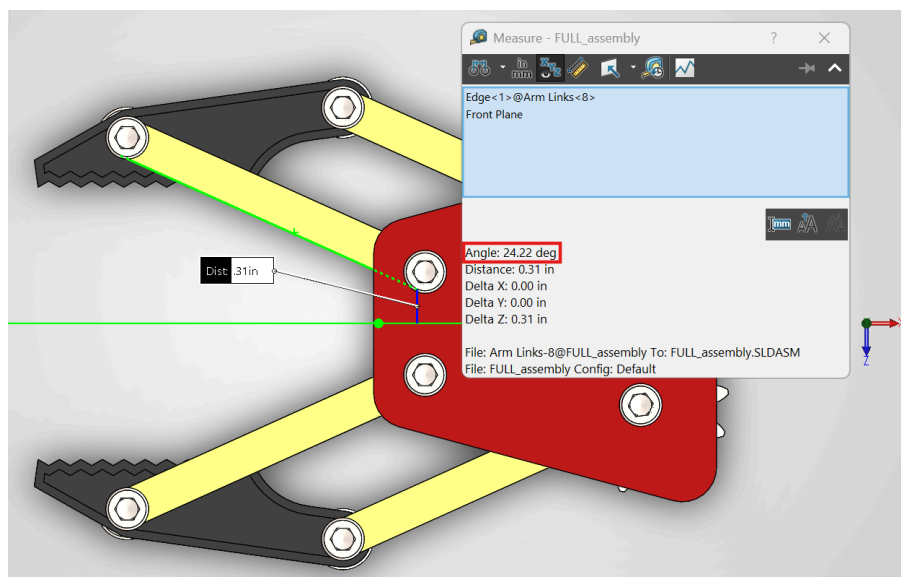


Figure 8. Open Claw Mechanism with Link 2 Angle

From Figure 8, we can see that when the mechanism is fully open link 2 makes a 24.22 degree angle with the negative global X axis.

The angle relative to the positive global X axis ($\theta_{2_{min}}$) can be determined,

$$\theta_{2_{min}} = 180^\circ - 24.22^\circ$$

$$\theta_{2_{min}} = 155.78^\circ$$

$$\text{Rounding} \rightarrow \theta_{2_{min}} = 156^\circ$$

From the calculations of $\theta_{2_{max}}$ and $\theta_{2_{min}}$, we have determined the range of θ_2 values to simulate our mechanism in MATLAB and Simulink.

MATLAB and Simulink - Simulation and Data

The following plots were obtained from MATLAB and Simulink:

(See [Appendix D](#) for MATLAB script and Simulink model)

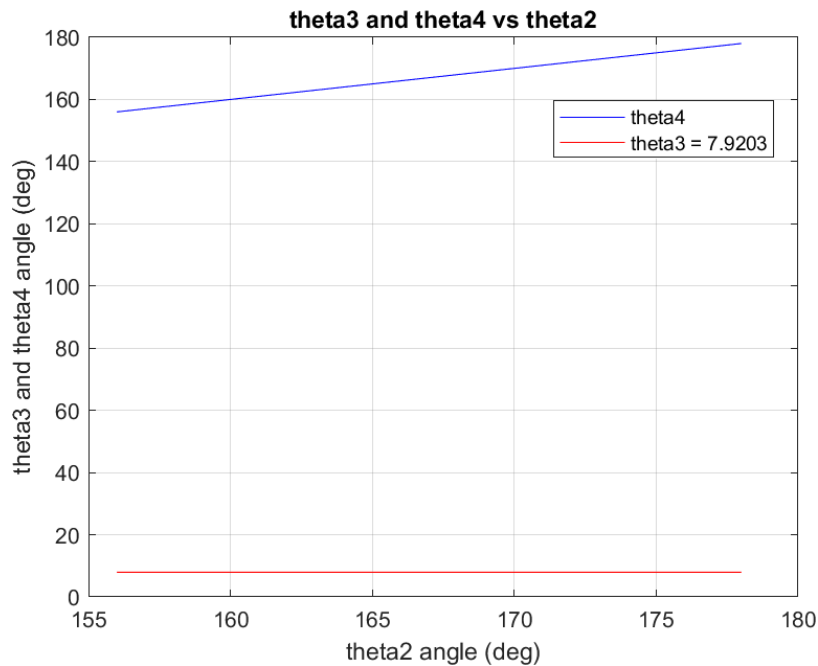


Figure 9. Theta3 and Theta4 vs Theta2 MATLAB Plot

Simple MATLAB plotting was used to plot theta3 and theta4 relative to theta2

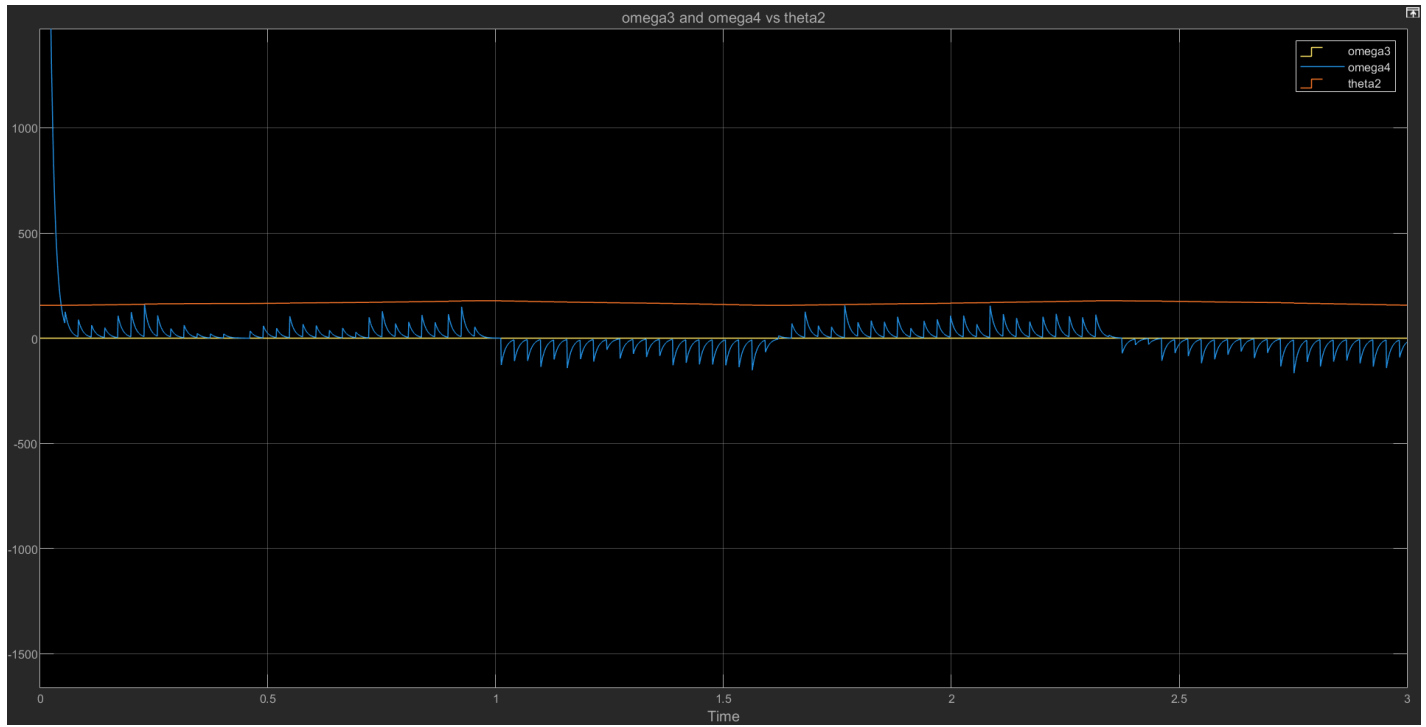


Figure 10. Omega3 and Omega4 vs Time Simulink Plot

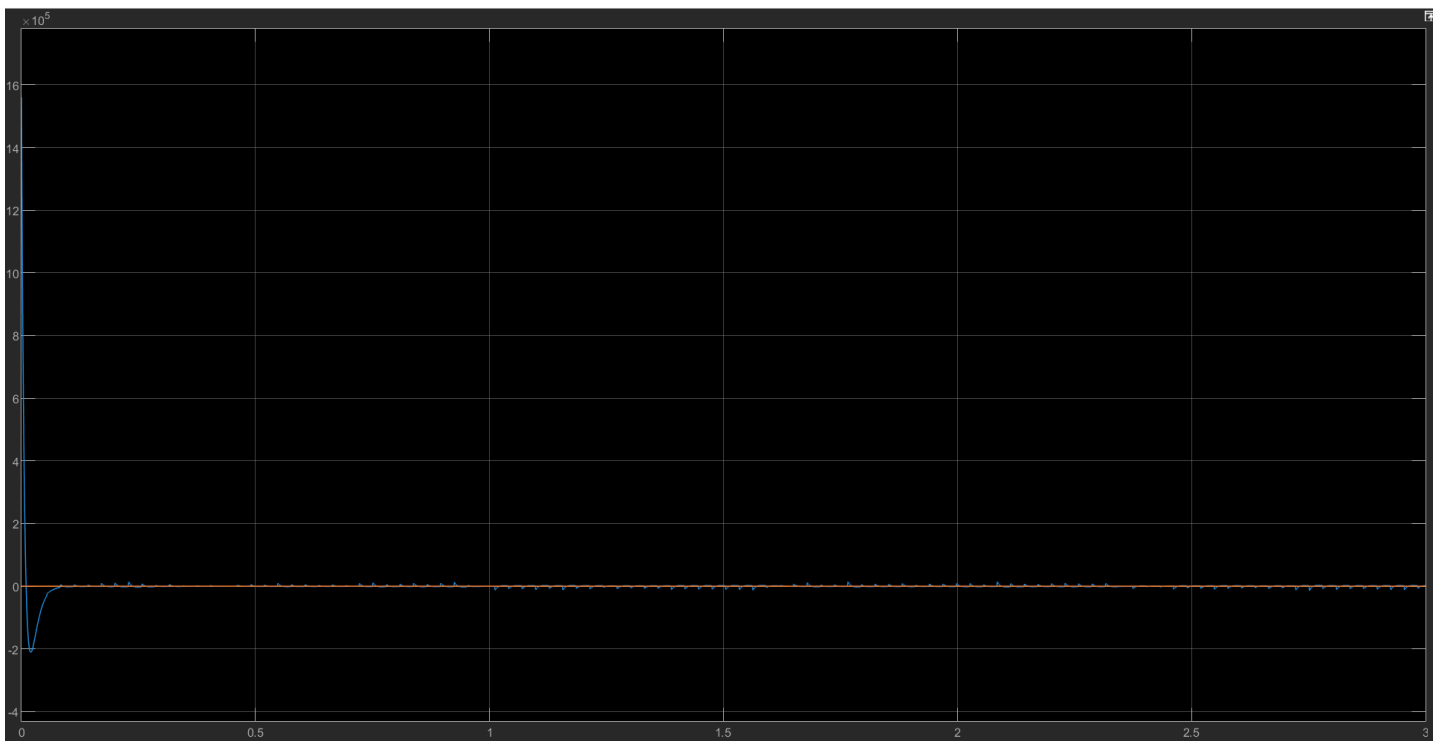


Figure 11. Alpha3 and Alpha4 vs Time Simulink Plot

The Simulink model was used to plot omega3 and omega4 relative to theta2, as well as plot alpha3 and alpha4 relative to theta2.

In Figure 10, the plot for omega4 appears spiky/jittery because it takes values directly from the slider block which represents theta2. When moving the slider block, there is input lag between the actual value on the slider and the value registered to the constant block for theta2. Because of this the values “jump” from one point to another repeatedly, therefore when taking the derivative with respect to time it results in a jittery plot. However, from the shape of the plot we can see that the velocity remains at a similar positive value when increasing theta2 and the same negative value when decreasing it.

In Figure 11, the plot for alpha4 is barely visible and remains close to 0 (apart from the startup acceleration). This is because the slider was moving at a relatively constant velocity.

Force Analysis

Additional Assumptions:

1. The input link rotates at constant angular speed, so $\alpha_2 = \alpha_4 = 0$
2. The claw is treated as part of the output member, or at minimum its payload force is applied as an external load on link 4.

The claw motion is symmetric, and the load is shared equally, so we will analyze one side only. These come from our report geometry, where the ground and coupler are 2 in, input and output are 3 in:

$$r_1 = 50.8 \text{ mm} = 0.0508 \text{ m}$$

$$r_2 = 76.2 \text{ mm} = 0.0762 \text{ m}$$

$$r_3 = 50.8 \text{ mm} = 0.0508 \text{ m}$$

$$r_4 = 76.2 \text{ mm} = 0.0762 \text{ m}$$

Masses of links: $m_2 = m_3 = m_4 = 3.53 \text{ g} = 0.00353 \text{ kg}$

Mass of claws: $m_c = 7.46 \text{ g} = 0.00746 \text{ kg}$

Then the object load on one finger is:

$$P = \frac{0.300 \times 9.81}{2} = 1.4715 \text{ N}$$

The contact distance from the output pivot is:

$$r_c = 5.05 \text{ in} = 0.12827 \text{ m}$$

Now, if we take the load as approximately perpendicular to the claw, the external moment from the object on one finger is:

$$M_p = Pr_c = 1.4715(0.12827) = 0.1887 \text{ N}\cdot\text{m}$$

We also know from our previous findings from our mechanism analysis for position, velocity, and acceleration, our special case four-bar linkage gives us the exact relations that simplify your force analysis a lot. We know:

$$\theta_4 = \theta_2$$

$$\theta_3 = \theta_1 = 0.138 \text{ rad}$$

$$\omega_4 = \omega_2$$

$$\omega_3 = 0$$

$$\alpha_3 = 0$$

$$\alpha_2 = \alpha_4 = 0$$

Next, we can find our center of gravity locations and inertias by using the common centerline assumption for the links:

$$b_2 = \frac{r_2}{2} = 0.0381 \text{ m}$$

$$b_3 = \frac{r_3}{2} = 0.0254 \text{ m}$$

$$b_4 = \frac{r_4}{2} = 0.0381 \text{ m}$$

$$\phi_2 = \phi_3 = \phi_4 = 0$$

For the link inertia, we know from our SOLIDWORKS CAD mass properties:

$$I_{G,link} = 1817.52 \text{ g}\cdot\text{mm}^2 = 1.8175 \times 10^{-6} \text{ kg}\cdot\text{m}^2$$

For the claw, we also know:

$$I_{G,claw} = 3293.76 \text{ g}\cdot\text{mm}^2 = 3.2938 \times 10^{-6} \text{ kg}\cdot\text{m}^2$$

Next, we can find our center of gravity accelerations by using the textbook's four-bar acceleration equations. For link 2, the chapter gives:

$$a_{G2x} = -b_2 \omega_2^2 \cos(\theta_2 + \phi_2)$$

$$a_{G2y} = -b_2 \omega_2^2 \sin(\theta_2 + \phi_2)$$

For link 3, the textbook expression simplifies a lot for your case because $\omega_3 = 0$ and $\alpha_3 = 0$. Then:

$$a_{G3x} = -r_2 \omega_2^2 \cos\theta_2$$

$$a_{G3y} = -r_2 \omega_2^2 \sin\theta_2$$

For link 4, with $\alpha_4 = 0$:

$$a_{G4x} = -b_4 \omega_2^2 \cos(\theta_2 + \phi_4)$$

$$a_{G4y} = -b_4 \omega_2^2 \sin(\theta_2 + \phi_4)$$

All of these come directly from the chapter 9 four-bar force-analysis setup.

Link 2: (Refer to [Appendix A.3 - Free Body Diagram of Link 2](#))

$$B_x - O_{2x} = m_2 a_{G2x}$$

$$B_y - O_{2y} = m_2 a_{G2y}$$

Moment about O_2 :

$$-r_2 \sin\theta_2 B_x + r_2 \cos\theta_2 B_y + M_{12} = I_{O2,2} \alpha_2$$

If constant speed is assumed, then $\alpha_2 = 0$, so this becomes

$$-r_2 \sin\theta_2 B_x + r_2 \cos\theta_2 B_y + M_{12} = 0$$

Link 3: (Refer to [Appendix A.4 - Free Body Diagram of Link 3](#))

$$B_x - C_x = m_3 a_{G3x}$$

$$B_y - C_y = m_3 a_{G3y}$$

Moment about G_3 :

$$-b_3 \sin \theta_3 B_x + b_3 \cos \theta_3 B_y - (r_3 - b_3) \sin \theta_3 C_x + (r_3 - b_3) \cos \theta_3 C_y = I_{G3} \alpha_3$$

Since $\alpha_3 = 0$:

$$-b_3 \sin \theta_3 B_x + b_3 \cos \theta_3 B_y - (r_3 - b_3) \sin \theta_3 C_x + (r_3 - b_3) \cos \theta_3 C_y = 0$$

Link 4 + claw: (Refer to [Appendix A.5 - Free Body Diagram of Link 4 and Claw](#))

$$C_x - O_{4x} + P_x = m_4 a_{G4x}$$

$$C_{34y} - O_{4y} + P_y = m_4^* a_{G4y}$$

The moment about O_4 is easiest because the claw load acts at a known distance from that pivot:

$$-r_4 \sin \theta_4 C_x + r_4 \cos \theta_4 C_y + M_P = I_{O4,4}^* \alpha_4$$

If constant speed is assumed, $\alpha_4 = 0$, so:

$$-r_4 \sin \theta_4 C_x + r_4 \cos \theta_4 C_y + M_P = 0$$

Matrix form

$$\{x\} = [O_{2x}, O_{2y}, B_x, B_y, C_x, C_y, O_{4x}, O_{4y}, M_{12}]^T$$

Now we know we need it in the form:

$$[A]\{x\} = \{B\}$$

$$\{B\} = \begin{bmatrix} m_2 a_{G2x} \\ m_2 a_{G2y} \\ I_{O2,2} \alpha_2 \\ m_3 a_{G3x} \\ m_3 a_{G3y} \\ I_{G3} \alpha_3 \\ m_4^* a_{G4x} - P_x \\ m_4^* a_{G4y} - P_y \\ I_{O4,4}^* \alpha_4 - M_P \end{bmatrix}$$

If constant speed is used, then the moment terms with $\alpha_2, \alpha_3, \alpha_4$ all become zero.

Matlab Implementation

The MATLAB code evaluates the dynamic force behavior of the gripper by sweeping the input angle θ_2 from 156° to 178° . For each position, it computes the accelerations of the link centers of mass and constructs a system of equilibrium equations in matrix form $[A]\{x\}=\{B\}$. The model includes the effect of the external load applied at the claw tip and accounts for the combined mass of the output link and gripper. See [Appendix D](#) for the script code.

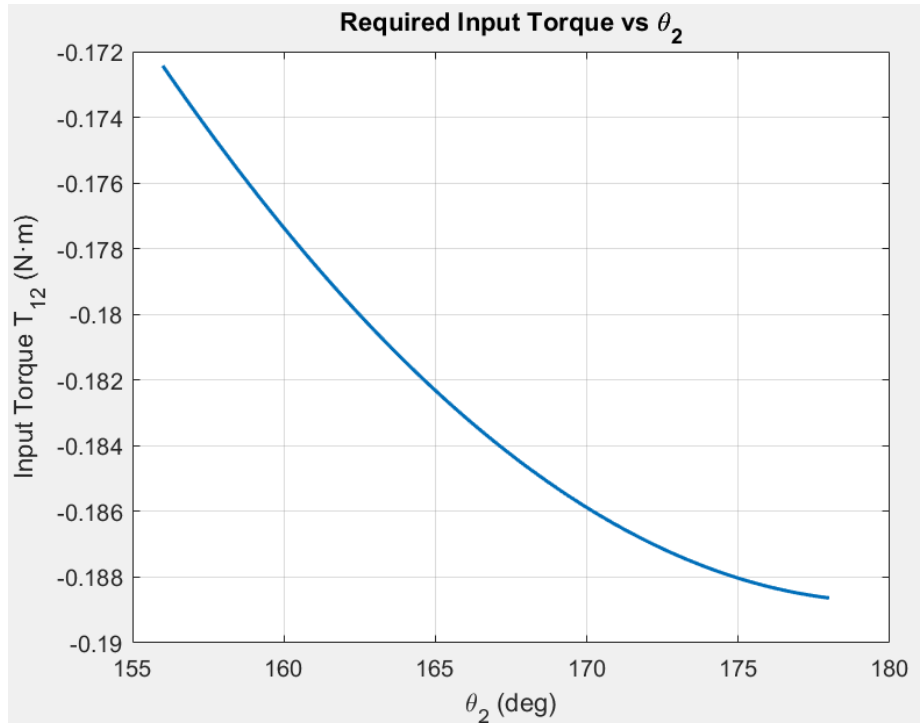


Figure 12. Required Input Torque vs Theta2

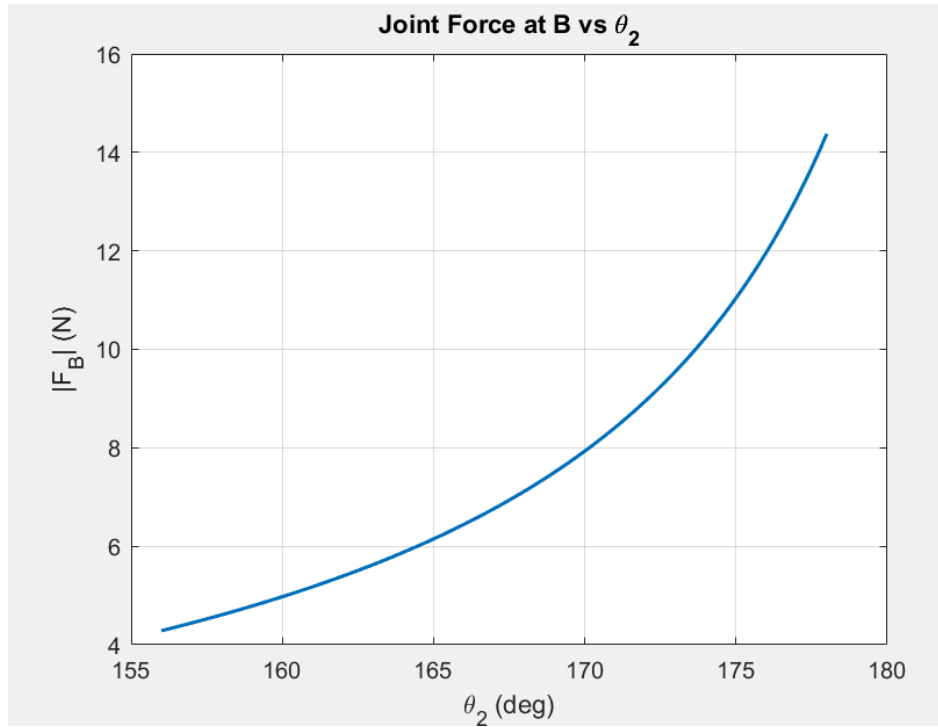


Figure 13. Joint Force at B vs Theta2

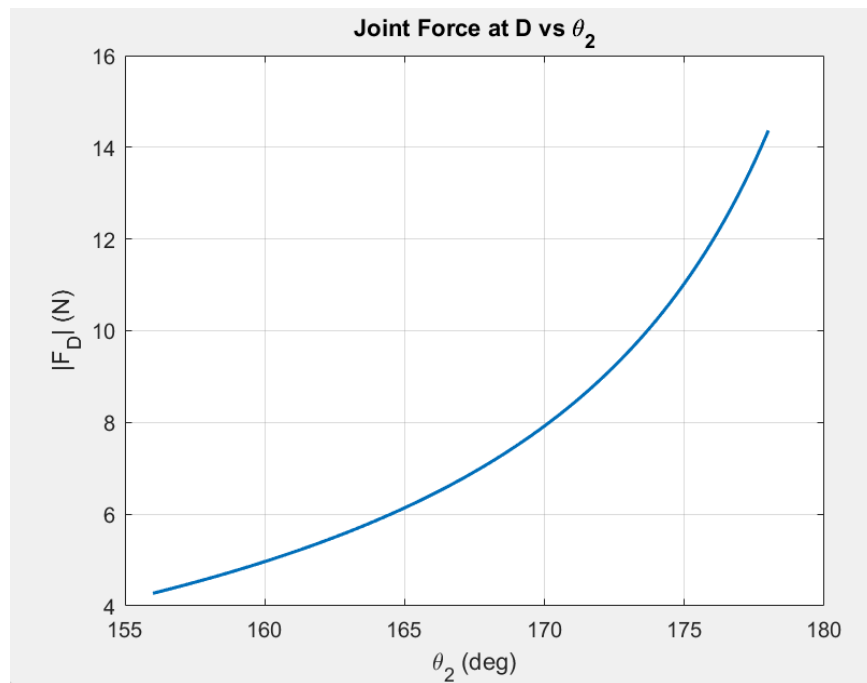


Figure 14. Joint Force at D vs Theta2

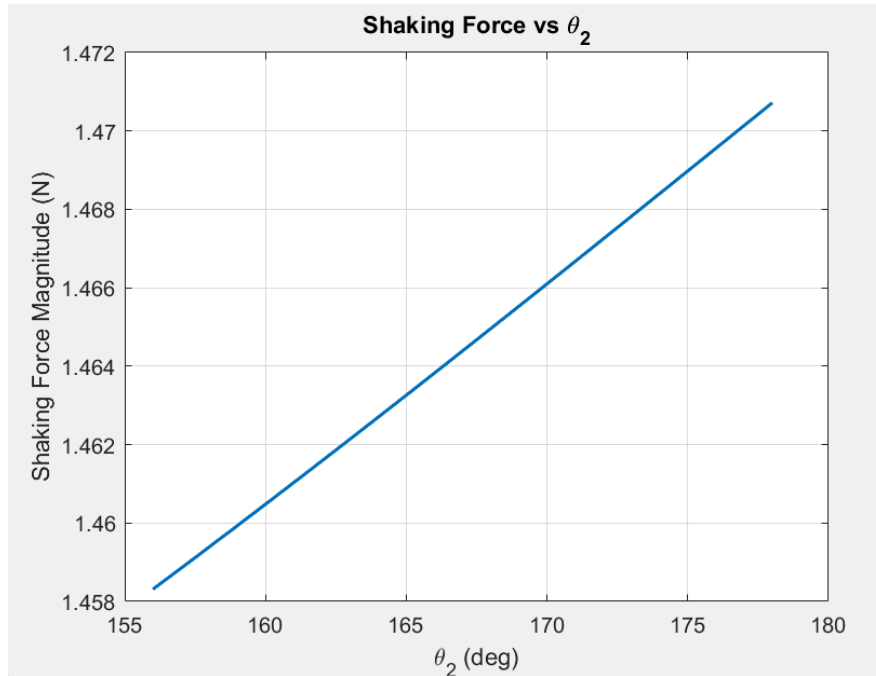


Figure 15. Shaking Force vs Theta2

The resulting graphs show how the system responds over the motion range. The input torque plot represents the torque required from the motor to drive the mechanism, while the joint force plots (FB and FD) show the internal forces at key joints. The shaking force plot represents the net reaction transmitted to the ground, indicating how much vibration or disturbance the system produces. Together, these plots help identify peak loads and the most critical operating positions of the mechanism.

ELECTRICAL SYSTEMS DESIGN METHODOLOGY

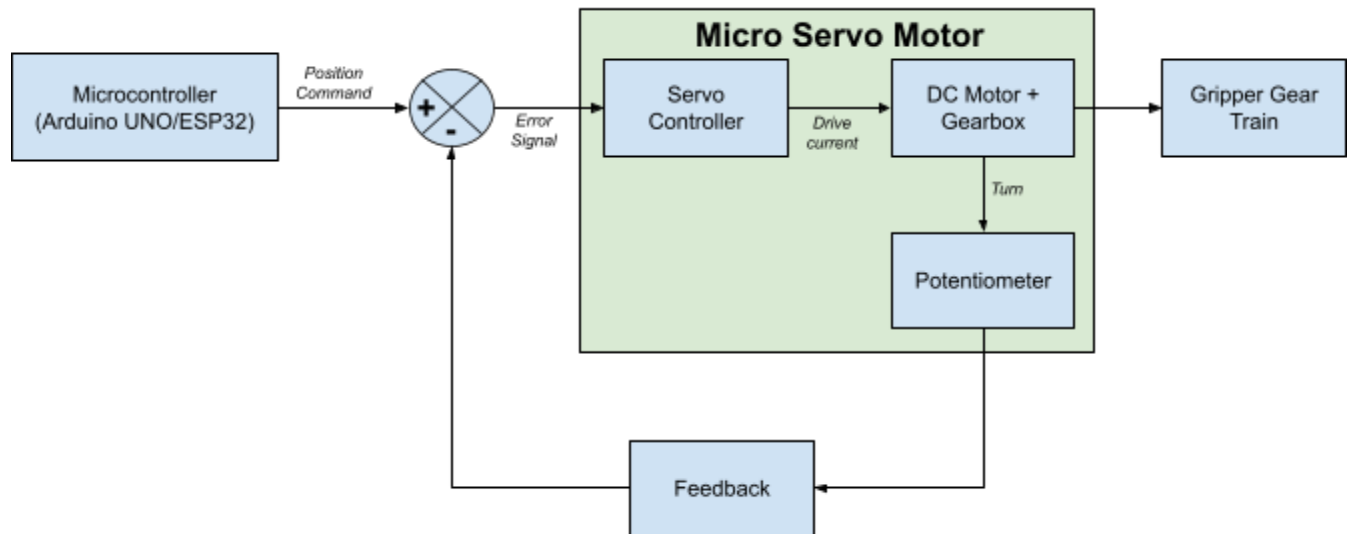


Figure 16. Electrical Systems Block Diagram

To clarify the electrical system the design does not use a servo motor to drive a separate DC motor. The servo motor itself already contains a small DC motor, an internal controller, a gearbox, and a potentiometer for position feedback. In our design, the microcontroller only sends a PWM position command to the servo, and then the servo's internal controller drives its own internal DC motor until the shaft reaches the commanded angle using feedback from the built-in potentiometer. So, the motor type is a micro servo motor, and the sensor type is the servo's internal potentiometer, which is why no separate external position sensor was specified in this design.

Motor Type

The gripper requires moderately high torque to be able to lift a minimum of 300g, it does not need high RPM. It is therefore evident that the type of motor required for this is some sort of DC motor. The gripper also requires highly accurate angular positioning and velocity control. Therefore, the specialty motor best suited for this gripper is a **servo motor**.

Sensors

Because a servo motor has an internal motor controller (sends appropriate current to the DC motor) and a potentiometer (measures angular position), no external sensors or peripherals are required.

Parts to be 3D Printed

1. Gripper fingers
2. Gear train housing and mounts
3. Servo mounting bracket
4. Base/frame of gripper

Brief Assembly Steps

1. Mount the servo to the 3D printed bracket
2. Attach the gear train to the servo shaft
3. Connect the gear train to the gripper fingers
4. Fix everything onto the base frame
5. Wire servo to microcontroller (power + PWM)

Estimated Cost

1. Servo motor: \$5–\$15
2. Microcontroller (Arduino/ESP32): \$10–\$20
3. Sensors: \$2–\$10
4. 3D printing material: \$5–\$15
5. Miscellaneous (wires, fasteners): \$5

Total Cost ~\$30 - \$60

Tests to be Performed

1. Testing the position accuracy. For example, command vs actual angle
2. Testing the gripping force, so whether it can hold an object
3. Finding the Response Time and Settling Time
4. Repeatability test at the same position multiple times
5. Load test, which is the max weight before failure

DISCUSSION

The designed mechatronic gripper has utilized multiple areas of focus, calculations, and theory. Combining these areas of focus has resulted in a complete design solution capable of performing the given tasks. The gripper has met the following constraints: it is a planar four-bar linkage, motion is completed within a 3-second operation cycle, it can produce enough force to grasp an object of 300g, it is compatible with rotary actuation, and has a gripper length of 6.5 inches when closed alongside an open finger distance of 2.5 inches. Furthermore, the mechanism demonstrated theoretical consistency with kinematic relationships through the use of MATLAB and Simulink modelling. Results confirmed the expected parallelogram behaviour through all analytical methods.

Understanding the kinematic results is essential for evaluating the design's effectiveness. The kinematic results may be divided into two significant parts: the positional analyses and the force analysis. The positional analyses focus on deriving position, velocity, and acceleration equations for the gripper. Final values are represented in angular terms, due to the gripper's rotational actuation. An important distinction when analyzing results is the omission of friction. This is due to the ambiguity of material selection in the final model. Despite its omission, the final results remain accurate for a general use case. Friction calculations would prevent clarity and simplified equations, while ultimately providing a negligible difference in perceived output. Positional results indicate that: $\theta_2 = \theta_4$ and $\theta_1 = \theta_3 = 7.92^\circ = 0.138rad$. θ_2 and θ_4 represent the input and output links, respectively. These values vary with the gripper's rotary actuation and thus are used to derive angular velocity and acceleration. The equal link angles confirm the parallelogram configuration of the mechanism. θ_1 and θ_3 represent the base angle and coupler angle, respectively. This allows for a constant value that can be solved. The angles being equal further confirm the parallelogram four-bar solution. The force calculations depend heavily on the results of the positional analysis. Furthermore, the assumption of no friction in the positional analysis may serve as a source of error for force calculations in a manufactured context. Alongside MATLAB, the results of the force analysis demonstrate that the gripper's torque and joint force reach a maximum as the input angle reaches its maximum. Through further calculation, it is demonstrated that the output force generated by the joints exceed the force required to grab the weight.

CONCLUSION

Iteratively designing, analyzing, and modelling a mechatronic gripper requires multiple disciplines of engineering to come together. Through kinetic analysis, dynamic analysis, MATLAB simulation, and electrical system design the gripper is designed to absolute perfection. Following the simulation, the desired design limitations are met in their entirety. The gripper has met the following constraints: it is a planar four-bar linkage, motion is completed within a 3-second operation cycle, it can produce enough force to grasp an object of 300g, it is compatible with rotary actuation, and has a gripper length of 6.5 inches when closed alongside an open finger distance of 2.5 inches. Combining all of the requirements with the mathematical models produces the finalized product. The gripper's finalized CAD models and geometric designs allow for a manufacturing ready mechatronics gripper. The project's purpose of designing and validating a mechatronic robotic gripper using a planar four bar linkage mechanism was completed with success.

FUTURE WORK

The current design & analysis of the mechatronic robotic gripper establishes a comprehensive theoretical and numerical foundation for a full implementation and fabrication of the gripper. To build upon these results, several areas for physical implementation, advanced software integration, as well as refined simulation and modeling have been identified and analyzed.

Enhancement of Theoretical & Simulation Frameworks

The current methodology of Simulated results can be enhanced to include more models for analysis. The effects of friction should be investigated in the MATLAB numerical model, which previously neglected friction to simplify the initial design.

Furthermore, rather than analyzing only one side of the gripper due to symmetry, a full system analysis could be conducted to account for potential torsional imbalances, effect of gears, as well as manufacturing errors.

While the parallelogram mechanism was selected through a subsystem matrix evaluation, future studies could utilize analytical synthesis to further optimize the link lengths for a better mechanical advantage.

A more complex consideration is to consider topology optimization, applying it to the 3D printed components to reduce overall mass without compromising structural integrity.

Physical Fabrication & Experimental Validation

While the current work relies solely on SolidWorks models and MATLAB simulations, the immediate next step is the fabrication of the gripper for real-world testing and validation. Utilizing the proposed manufacturing plan, the base plate, linkages and gripper fingers should be 3D printed for rapid prototyping and verification of the assembly.

In the Electrical Design Methodology, it highlights the tests that should be performed, including repeatability , response time measurements, and load-failure testing for real world validation.

Furthermore, future work should move beyond the assumptions, incorporating and investigating friction, tolerances, and material deformation further.

Advanced Control Systems

Future iterations should implement real-time software control rather than relying on conceptual integration. Implementing a robust closed-loop control system would allow for precise positioning and force regulation in real-time, this moves beyond the servo's internal potentiometer and creates a more advanced control system, both PI and PID controllers should be investigated for future applications.

To improve grasping reliability, external sensors such as tactile pressure sensors should be included in the implementation and fabrication of the gripper. This would allow the system to detect slip and also understand the grasping force necessary for a specific object, as well as detecting slip. Furthermore, to improve the jittery data spikes from the simulink model caused by the manual slider input lag, proposing the usage of filters such as Low-pass or Band-pass filters should be investigated. Further improvement of the simulink environment can be accomplished by integrating the physical motor and sensors within the environment, allowing for more accurate dynamic response.

System Integration & Applications

The gripper is designed to be an end-effector, and its utilization is best demonstrated through higher-level integration. Future work includes mounting the gripper onto a multi-DOF robotic arm, allowing for further testing and validation of complex operations that are standard in industrial applications or other engineering environments.

Integrating a camera system would enable target detection and the software could calculate the required θ_2 ranges for various object trajectories, automating the grippers function.

Overall, these considerations will ensure the gripper evolves from a validated analytical concept into a robust intelligent tool suitable for a diverse range of applications.

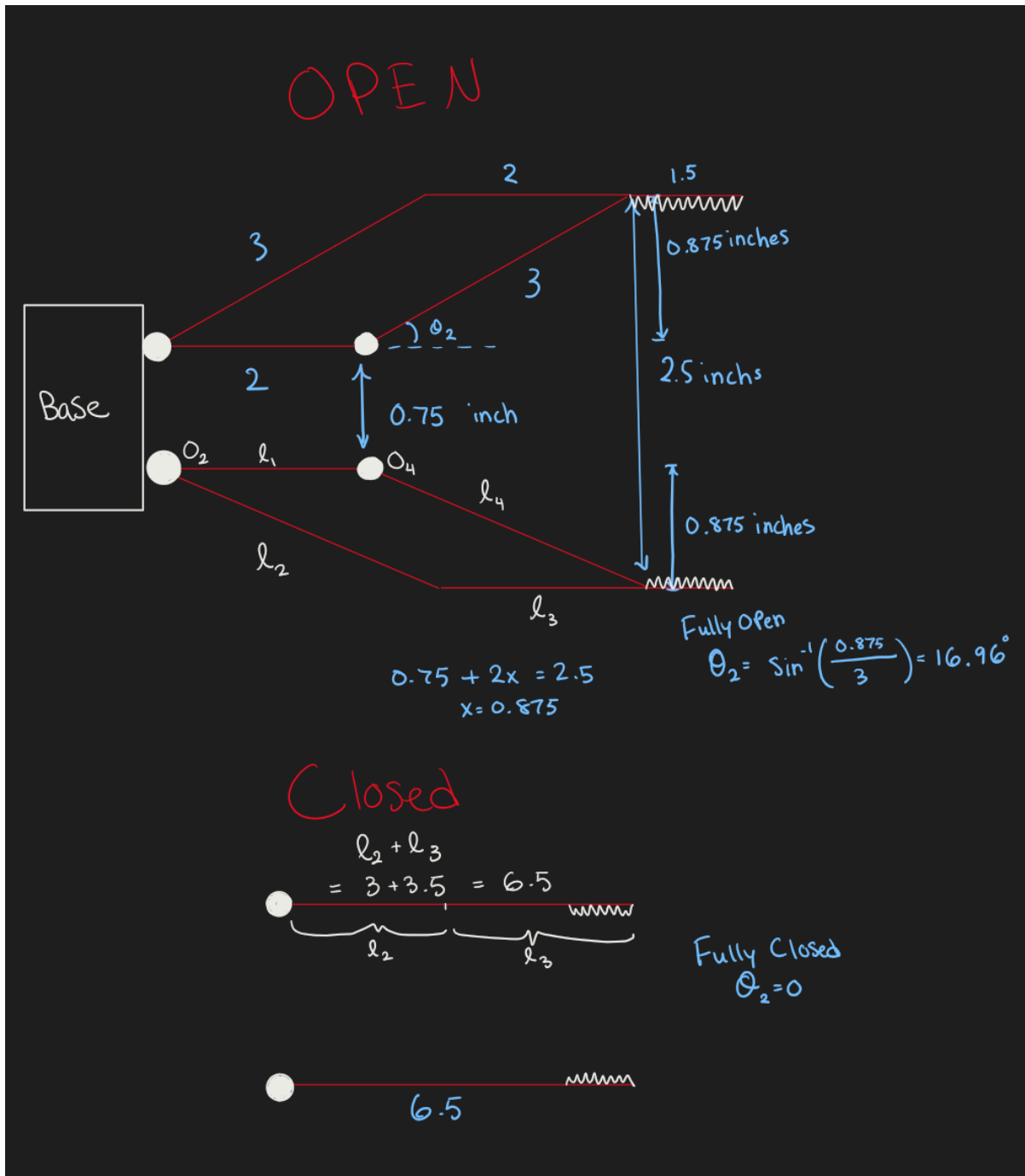
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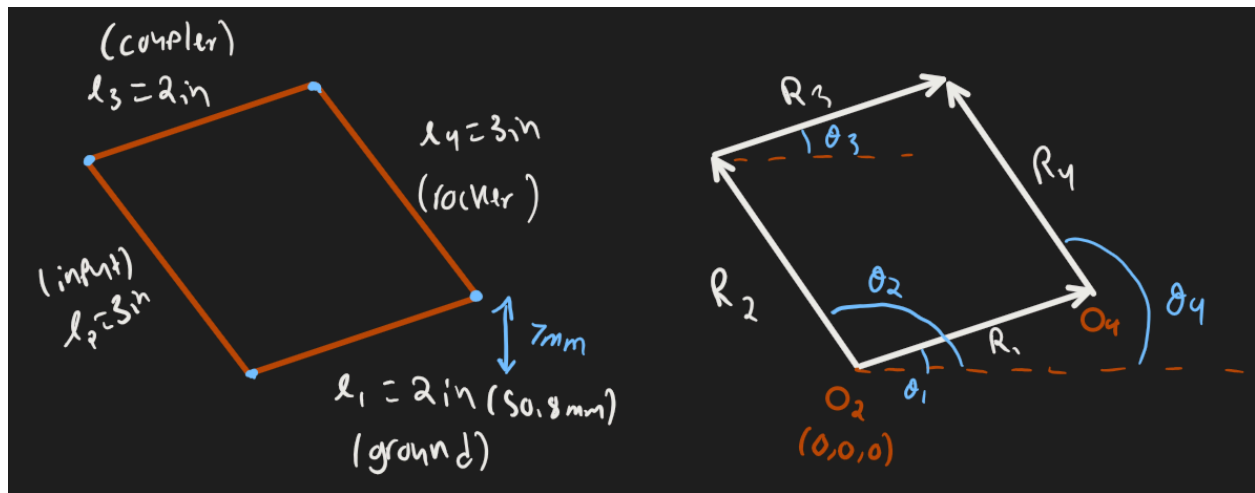
APPENDICES

Appendix A - Design Considerations

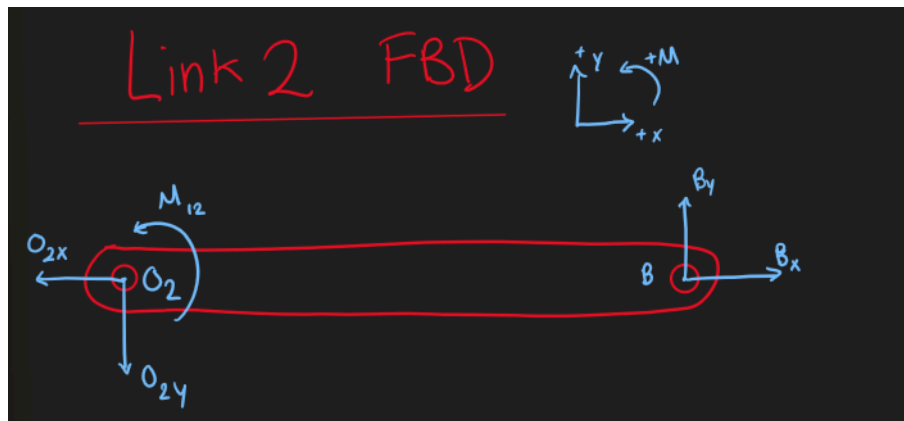
1. Graphical Synthesis Calculations



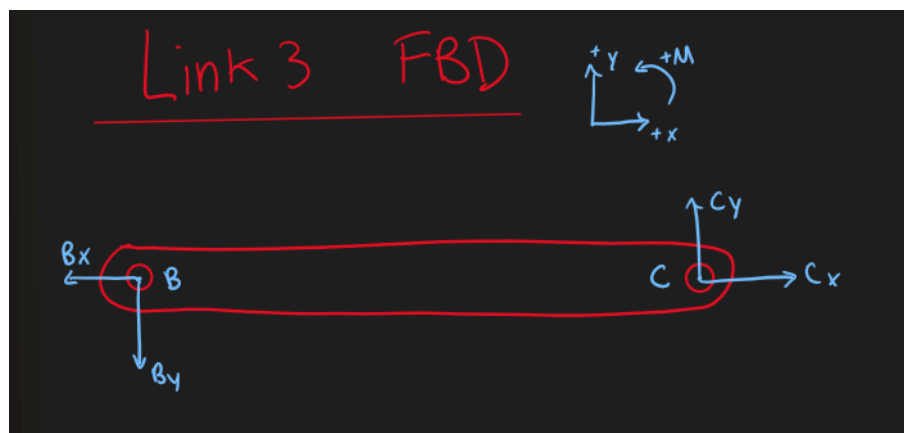
2. Assembly Configuration and Setup



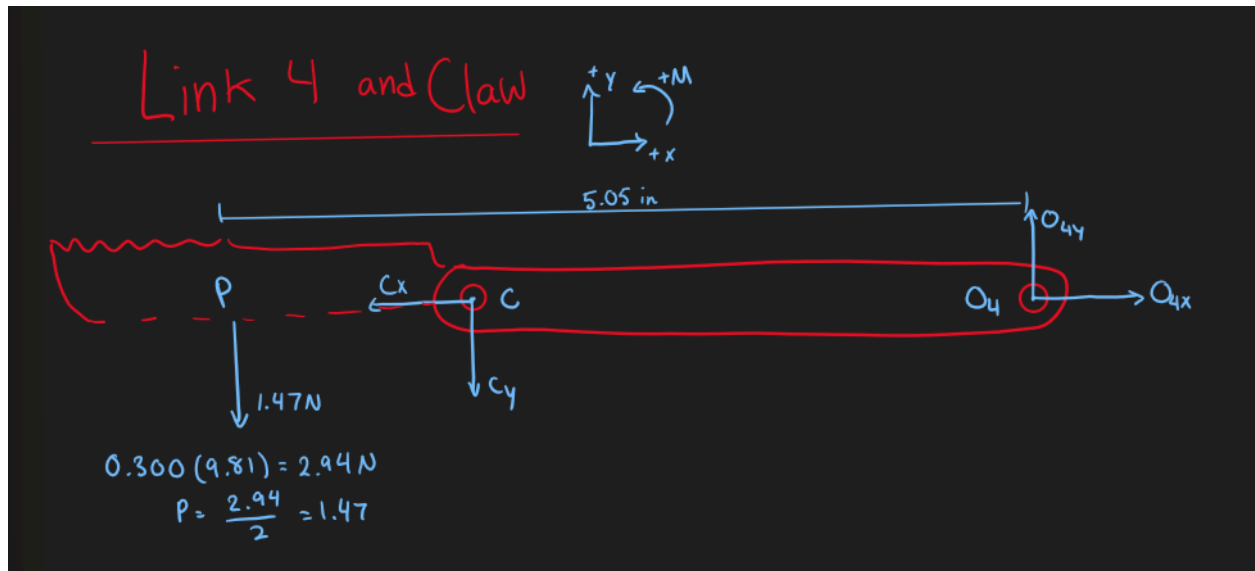
3. Free Body Diagram of Link 2



4. Free Body Diagram of Link 3

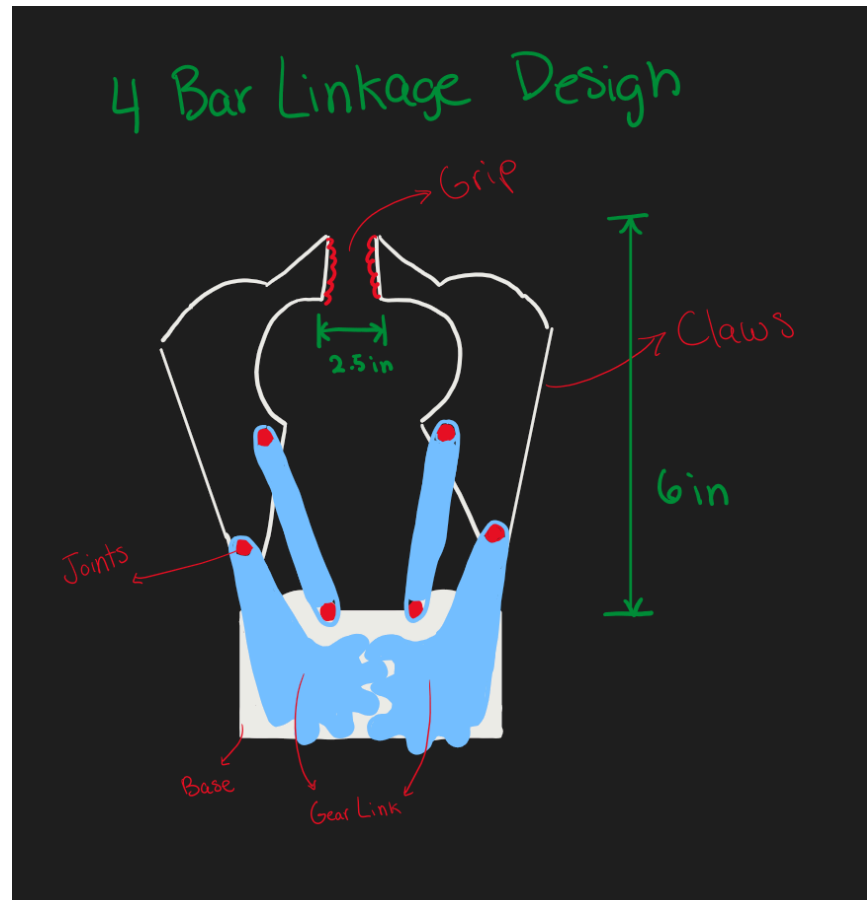


5. Free Body Diagram of Link 4 and Claw

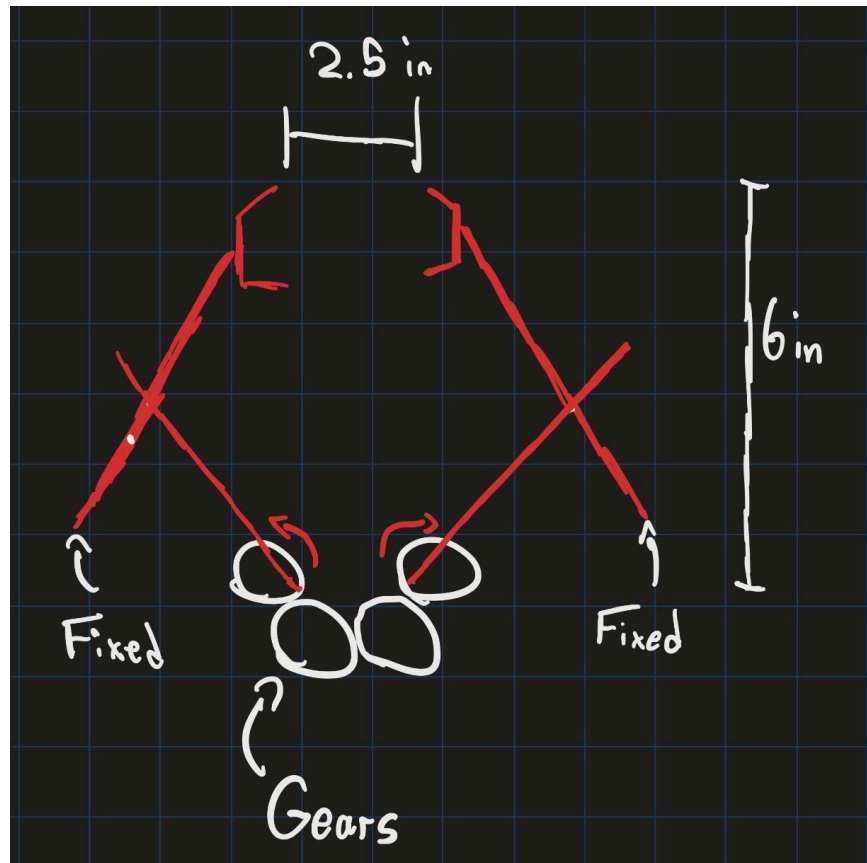


Appendix B - Design Concepts

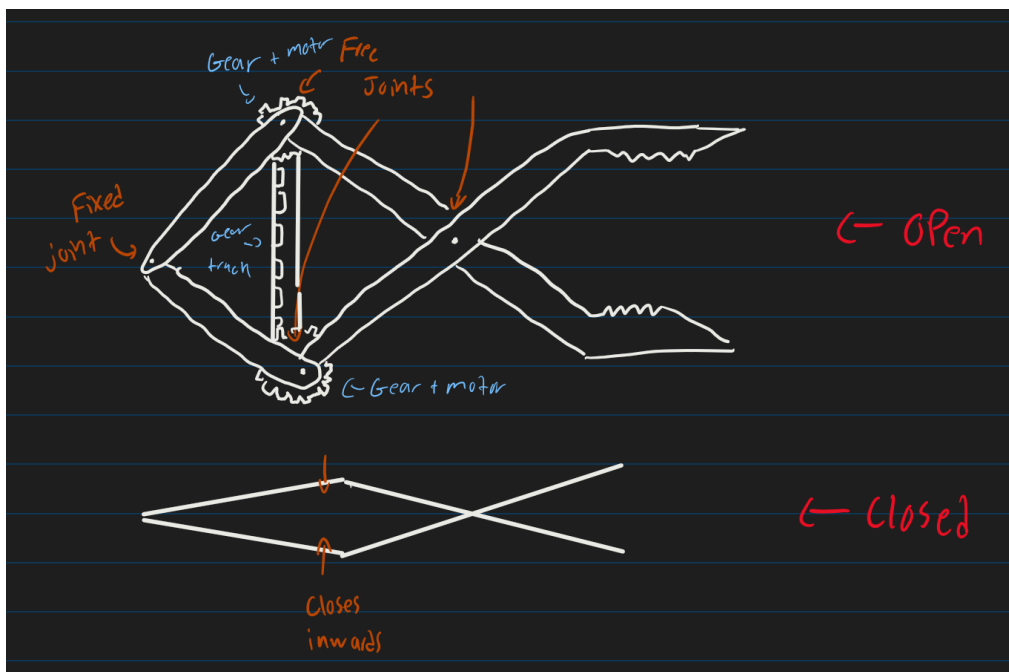
1. Design Concept 1



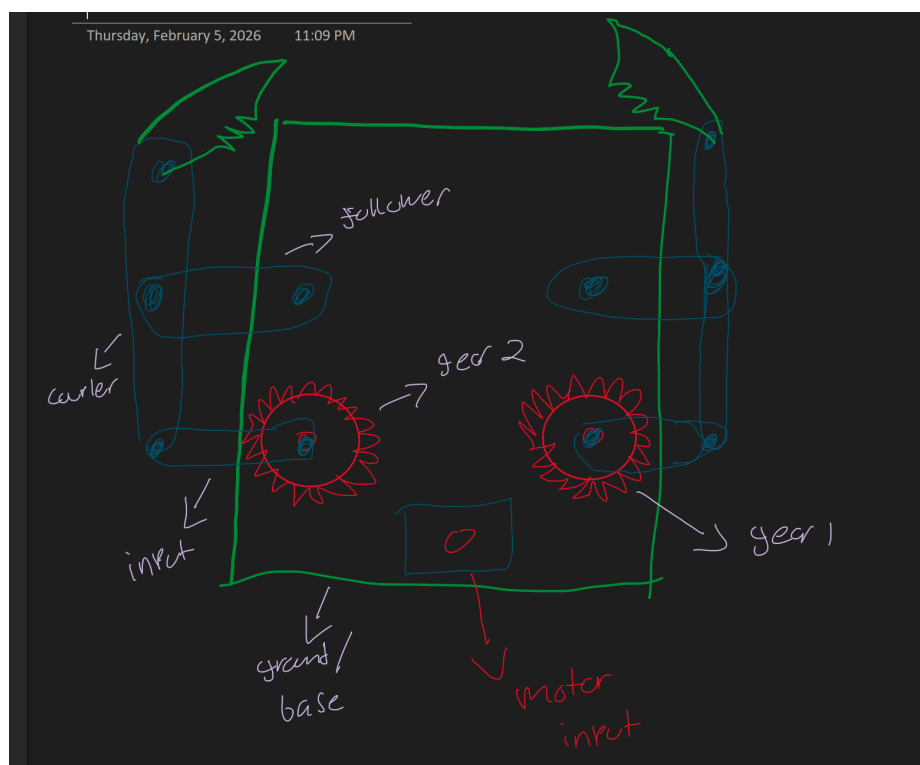
2. Design Concept 2



3. Design Concept 3



4. Design Concept 4



Appendix C - Mechanism Selection

1. Evaluation Scale

1	Poor
2	Below Average
3	Acceptable
4	Good
5	Excellent

2. Design Selection Chart

Design Objectives	Weight	Concept 1	Concept 2	Concept 3	Concept 4
Use of a planar four-bar linkage	5	5	1	5	5
Has assembled total length of 6.5 inches when closed	5	5	5	5	5
Distance between fingers when fully opened is 2.5 inches	5	5	5	5	5
Can be realistically implemented	4	5	3	3	5
Ease of rotary actuation integration	3	5	5	5	5
Smoothness of motion	3	4	2	2	4
Ease of modeling and analysis	3	4	2	2	4
Simplicity	3	3	3	2	3
Compactness and Symmetry	3	5	5	5	5
TOTAL (Sum of Weight x Score):	-	158	118	135	158

3. Final Choice

We created several conceptual four-bar linkage gripper configurations and explored pros and cons of each before finalizing the selected mechanism. Even though most of the designs cleared the requirement of using a four bar linkage, they all differed in link arrangement, transmission characteristics, force transfer, and ease of modeling on cad. After sifting through, we took the concepts that satisfied all requirements and compared them to the following criteria:

1. Use of a planar four-bar linkage
2. Ability to meet 2.5 in opening and 6.5 in closed reach
3. Realistic Implementation
4. Ease of rotary actuation integration
5. Smoothness of motion
6. Ease of analytical modeling and Simplicity
7. Compactness and symmetry

After it came down to choosing the final concepts, we took the best parts of all the concepts and created the best version of them all. We created the design so that the drag link motion allows continuous rotary actuation, which satisfies the project constraint requiring compatibility with rotary actuation. This would simplify the motor integration and get rid of the need for complex slider systems. Furthermore, we chose this design because the geometry produces smooth angular displacement of the output link, which would result in a controlled claw opening and closing sequence. Adding the gears would control the sudden acceleration spikes that we would get from a motor. Moreover, the selected link proportions increase the mechanical advantage as the claw would approach the closed position. This is beneficial because we will need the gripping force to be most crucial near full closure when holding a 300g weighted object. Also, this configuration that we created allows for a very straightforward implementation of MATLAB numerical simulation because of its simple geometry. To add, the symmetric claw design improves the load distribution and reduces any torsional imbalance during gripping. It also really simplifies the CAD assembly. To conclude, this four-bar claw mechanism that we chose is deliberately proportioned to satisfy both geometric and dynamic requirements.

Appendix D - Matlab Code & Simulink Model

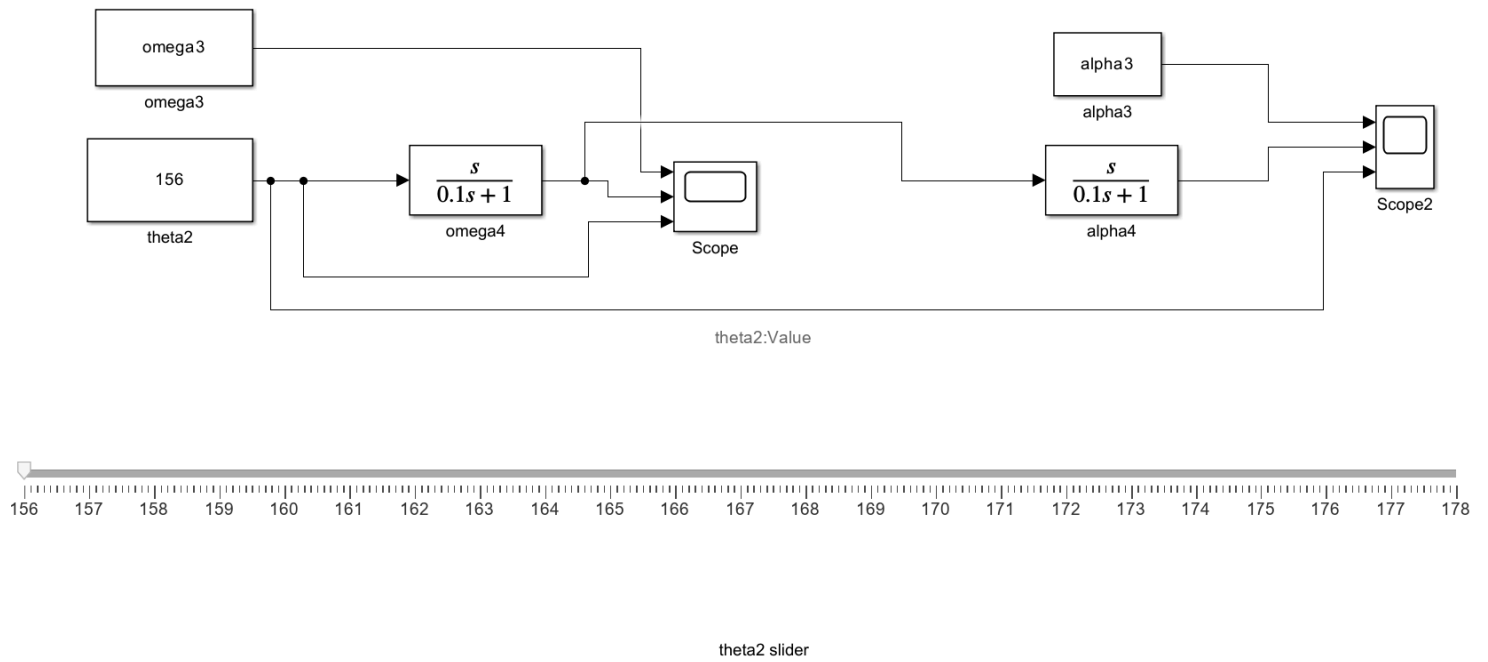
1. Matlab Code for Angular Displacement

```

clear;
clc;
%-----INITIALIZING-----
% link lengths in mm (50.8mm = 2in and 76.2mm = 3in)
l1 = 50.8;
l2 = 76.2;
l3 = 50.8;
l4 = 76.2;
% POSITION EQUATIONS:
% Fixed angles theta1 and theta3,
% 7 is the vertical offset in mm between O2 and O4
theta1 = asind(7/l1);
theta3 = asind(7/l3);
% Arrays of min-max angle values of theta2 and theta4
theta2 = 156:1:178;
theta4 = theta2;
% VELOCITY EQUATIONS:
syms x
f1 = sym(theta1);
f2 = sym(theta3);
omega1_val = diff(f1, x);
omega3_val = diff(f2, x);
omega1 = double(omega1_val);
omega3 = double(omega3_val);
% ACCELERATION EQUATIONS:
f3 = sym(omega1);
f4 = sym(omega3);
alpha1_val = diff(f3, x);
alpha3_val = diff(f4, x);
alpha1 = double(alpha1_val);
alpha3 = double(alpha3_val);
%-----PLOTTING-----
% Plotting theta3 and theta4 vs theta2 graph:
plot(theta2, theta4, 'b');
title('theta3 and theta4 vs theta2');
xlabel('theta2 angle (deg)');
ylabel('theta3 and theta4 angle (deg)');
grid on;
hold on;
theta3_plotting_array = theta3 * ones(size(theta2));
plot(theta2, theta3_plotting_array, 'r');
legend('theta4', ['theta3 = ', num2str(theta3)]);
hold off;

```

2. Simulink Model for Angular Velocity and Acceleration



3. Matlab Code for Force Analysis

```

clear; clc; close all;

r1 = 2.0 * 0.0254;
r2 = 3.0 * 0.0254;
r3 = 2.0 * 0.0254;
r4 = 3.0 * 0.0254;

theta3 = deg2rad(7.92);
theta2_vec = deg2rad(linspace(156,178,100));
theta4_vec = theta2_vec;

m2 = 3.53e-3;
m3 = 3.53e-3;
m4 = 3.53e-3;
m_claw = 7.46e-3;

m4_total = m4 + m_claw;

b2 = r2/2;
b3 = r3/2;
b4_eff = 0.0398;

W_total = 0.300 * 9.81;
P = W_total / 2;

Px = 0;
Py = -P;

rc = 5.05 * 0.0254;

omega2 = 2*pi;

alpha2 = 0;
alpha3 = 0;
alpha4 = 0;
omega3 = 0;
omega4 = omega2;

n = length(theta2_vec);

O2x = zeros(1,n);
O2y = zeros(1,n);
Bx = zeros(1,n);
By = zeros(1,n);
Dx = zeros(1,n);
Dy = zeros(1,n);
O4x = zeros(1,n);
O4y = zeros(1,n);

```

```

T12 = zeros(1,n);

F_B = zeros(1,n);
F_D = zeros(1,n);
F_shake = zeros(1,n);

for k = 1:n

    theta2 = theta2_vec(k);
    theta4 = theta4_vec(k);

    c2 = cos(theta2); s2 = sin(theta2);
    c3 = cos(theta3); s3 = sin(theta3);
    c4 = cos(theta4); s4 = sin(theta4);

    aG2x = -b2 * omega2^2 * c2;
    aG2y = -b2 * omega2^2 * s2;

    aG3x = -r2 * omega2^2 * c2;
    aG3y = -r2 * omega2^2 * s2;

    aG4x = -b4_eff * omega4^2 * c4;
    aG4y = -b4_eff * omega4^2 * s4;

    A = zeros(9,9);
    Bvec = zeros(9,1);

    A(1,:) = [1 0 1 0 0 0 0 0 0];
    Bvec(1) = m2 * aG2x;

    A(2,:) = [0 1 0 1 0 0 0 0 0];
    Bvec(2) = m2 * aG2y;

    A(3,:) = [0 0 -r2*s2 r2*c2 0 0 0 0 1];
    Bvec(3) = 0;

    A(4,:) = [0 0 -1 0 1 0 0 0 0];
    Bvec(4) = m3 * aG3x;

    A(5,:) = [0 0 0 -1 0 1 0 0 0];
    Bvec(5) = m3 * aG3y;

    A(6,:) = [0 0 -b3*s3 b3*c3 -b3*s3 b3*c3 0 0 0];
    Bvec(6) = 0;

    A(7,:) = [0 0 0 0 -1 0 1 0 0];
    Bvec(7) = m4_total * aG4x - Px;

    A(8,:) = [0 0 0 0 0 -1 0 1 0];

```

```

Bvec(8) = m4_total * aG4y - Py;

A(9,:) = [0 0 0 0 r4*s4 -r4*c4 0 0 0];
Bvec(9) = -(rc*c4*Py - rc*s4*Px);

x = A \ Bvec;

O2x(k) = x(1);
O2y(k) = x(2);
Bx(k) = x(3);
By(k) = x(4);
Dx(k) = x(5);
Dy(k) = x(6);
O4x(k) = x(7);
O4y(k) = x(8);
T12(k) = x(9);

F_B(k) = sqrt(Bx(k)^2 + By(k)^2);
F_D(k) = sqrt(Dx(k)^2 + Dy(k)^2);
F_shake(k) = sqrt((O2x(k)+O4x(k))^2 + (O2y(k)+O4y(k))^2);
end

theta2_deg = rad2deg(theta2_vec);

figure;
plot(theta2_deg, T12, 'LineWidth', 1.5);
grid on;
xlabel('\theta_2 (deg)');
ylabel('Input Torque T_{12} (N·m)');
title('Required Input Torque vs \theta_2');

figure;
plot(theta2_deg, F_B, 'LineWidth', 1.5);
grid on;
xlabel('\theta_2 (deg)');
ylabel('|F_B| (N)');
title('Joint Force at B vs \theta_2');

figure;
plot(theta2_deg, F_D, 'LineWidth', 1.5);
grid on;
xlabel('\theta_2 (deg)');
ylabel('|F_D| (N)');
title('Joint Force at D vs \theta_2');

figure;
plot(theta2_deg, F_shake, 'LineWidth', 1.5);
grid on;
xlabel('\theta_2 (deg)');

```

```
ylabel('Shaking Force Magnitude (N)');  
title('Shaking Force vs \theta_2');  
  
fprintf('Max input torque = %.5f N·m\n', max(abs(T12)));  
fprintf('Max joint force at B = %.5f N\n', max(F_B));  
fprintf('Max joint force at D = %.5f N\n', max(F_D));  
fprintf('Max shaking force = %.5f N\n', max(F_shake));
```